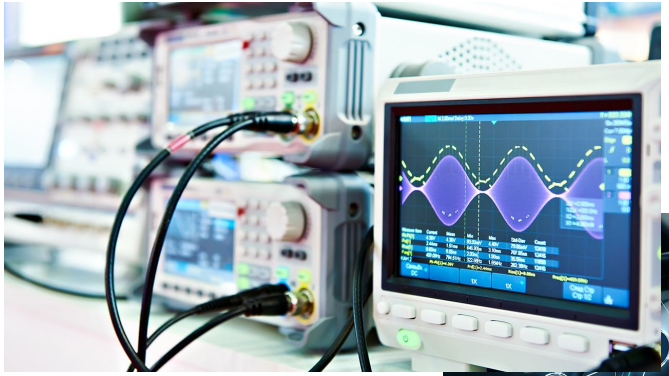
Bare-Metal OS for Raspberry Pi 4

Andrea Seehuber, Luca Fässler

Operating Systems FS 2026

University of Basel, 26.03.2026

Why design an own OS?



https://de.digi.com/

Embedded Devices:

- limited CPU and memory resources
- specific functionality
- no need for generic overhead
- usually RTOS

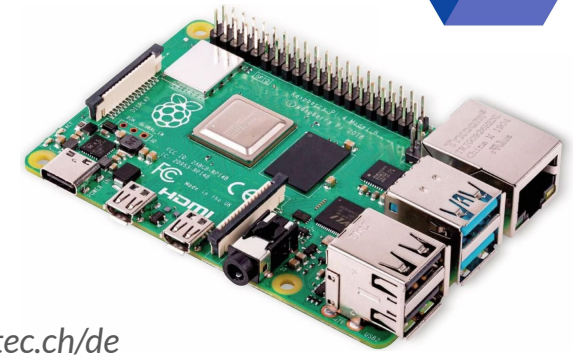
Objective

Design and implement a minimal bare-metal OS kernel for the Raspberry Pi 4 that demonstrates fundamental OS concepts, including:

- system bootstrapping
- low-level hardware interaction
- task management and scheduling
- interrupt handling
- time-based execution

Hardware and Developing Tools

- Raspberry Pi 4, ARM Cortex-A72
- UART serial interface for debugging and system interaction
- Cross-compiler: aarch64-none-elf-gcc
- Programming languages: C and a bit of Assembly
- Build system: Make
- Serial terminal (e.g. PuTTY)



https://www.digitec.ch/de

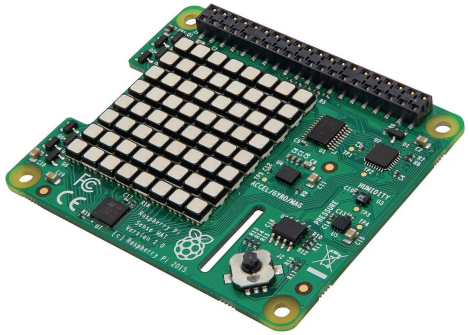
Core Components

- Boot and Kernel Initialization
  - Assembly Bootstrap
  - linker script
  - transition to kernel_main()
  - stack setup and early hardware init
- UART Driver
  - serial communication
  - basic console output
- Interrupt handling and timer
  - exception vector table
  - IRQ handling
  - hardware timer initialization
  - global tick counter
- Task System and Scheduler
  - Task Control Block (TCB)
  - context switching
  - Round-Robin scheduling (cooperative)

Task Design

- CPU-bound Task
  - continuously performs computations
  - explicitly yields control
  - demonstrates cooperative scheduling
- UART/Event Task
  - waits for user input via UART
  - becomes active only upon events
  - demonstrates blocking behaviour
- Timer-based Task
  - sleeps for a defined number of ticks
  - wakes up periodically and performs work
  - demonstrates time-based execution and wakeup
- Idle Task
  - runs when no other task is runnable
  - ensures scheduler always has a valid fallback
  - represents system idle state

Extensions



https://www.digitec.ch/de

Would be cool …

- I2C driver for Sense HAT communication
- VideoCore mailbox interface for HDMI output

Timeline

	23.03.	30.03.	06.04.	13.04.	20.04.	27.04.	04.05.	11.05.	18.05.	25.05.	01.06.	08.06.
Foundations			EASTER							OS EXAM		
Interrupts and Time												
Task System												
Task Behaviours												
Extensions												
Debugging and Docs												

Questions?